

Chaoyi Zhao 赵朝熠

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Affiliations	Massachusetts Institute of Technology, Sloan School of Management	MA, USA
	Postdoctoral Associate, MIT Laboratory for Financial Engineering	2024 – Present
	– Advisor: Prof. Andrew W. Lo	
	Society of Actuaries	
	Fellow (FSA, SOA)	2024 – Present
	Associate (ASA, SOA)	2023 – 2024
Research Interests	Investments: Portfolio theory, Factor investing, Market microstructure	
	Statistics: High-dimensional statistics, Machine learning, Artificial intelligence	
	Actuarial Sciences: Copula theory, Term structure modeling	
Education	Peking University, School of Mathematical Sciences	Beijing, China
	Ph.D. in Statistics	Sep 2019 – Jan 2024
	– Advisors: Prof. Lan Wu and Prof. Ruixun Zhang	
	– Completed six months ahead of schedule	
	Peking University, School of Mathematical Sciences	Beijing, China
	B.S. in Mathematics and Applied Mathematics	Sep 2015 – Jul 2019
	– Elite Undergraduate Program of Applied Mathematics	Sep 2017 – Jul 2019
	Peking University, School of EECS	Beijing, China
	B.S. (Double Degree) in Computer Science and Technology	Sep 2016 – Jul 2019
Honors	Awards	
	Outstanding Doctoral Dissertation Award (Peking University)	2024
	Beijing Outstanding Graduate	2019, 2024
	Excellent Graduate (Peking University)	2019, 2024
	Merit Student (Peking University)	2016, 2020, 2021, 2022, 2023
	S&P Global Academic ESG Research Award	2022
	Second place in the Best Paper Prize for Young Scholars, Annual Conference of the Operations Research Society of China (Financial Engineering and Risk Management Branch)	2022
	Excellent Graduate of the Elite Undergraduate Program of Applied Mathematics	2019
	The Student of The Year (10 students per year, Peking University)	2018
	Merit Student Pacesetter (Peking University)	2018
	Beijing Merit Student	2018
	Excellent Student Leader (Peking University)	2017

Scholarships

National Scholarship	2018, 2023
President Scholarship (Peking University)	2019, 2020, 2021, 2022
Leo KoGuan Scholarship (Peking University)	2022
Huatai Science and Technology Scholarship (Peking University)	2021
CETC the 14th Research Institute Glarun Scholarship (Peking University)	2020
The National Southwest Associated University Scholarship (Peking University)	2018
Shenzhen Finance Institute Scholarship (the Chinese University of Hong Kong, Shenzhen)	2017
Founder Scholarship (Peking University)	2016

Publications

Alphabetical order; * Corresponding author.

Portfolio Theory and Investing

1. The Health of Nations Fund: Financing Global Drug Development
Joonhyuk Cho, Manish Singh, Chaoyi Zhao, Shomesh E. Chaudhuri, and Andrew W. Lo.
PLOS Global Public Health, forthcoming, 2026.
2. High-Frequency Liquidity in the Chinese Stock Market: Measurements, Patterns, and Determinants
Chaoyi Zhao, Yufan Chen, Lintong Wu, Yuehao Dai, Ermo Chen, Lan Wu, and Ruixun Zhang.
Pacific-Basin Finance Journal, 90, 102681, 2025. [[PDF](#)][[Appendix](#)][[Journal](#)]
3. Optimal Impact Portfolios with General Dependence and Marginals
Andrew W. Lo, Lan Wu, Ruixun Zhang, and Chaoyi Zhao^{#*}.
Operations Research, 72(5), 1775–1789, 2024. [[PDF](#)][[Appendix](#)][[Journal](#)]
– Second place in the Best Paper Prize for Young Scholars, Annual Conference of the Operations Research Society of China (Financial Engineering and Risk Management Branch).
– Reported by [Center for Statistical Science](#), Peking University.
4. Measuring and Optimizing the Risk and Reward of Green Portfolios
Andrew W. Lo, Ruixun Zhang, and Chaoyi Zhao^{#*}.
The Journal of Impact and ESG Investing, 3(2), 55–93, 2022. [[PDF](#)][[Journal](#)]
– Winner of the [S&P Global Academic ESG Research Award](#).
– Reported by [Practical Applications](#) [[PDF](#)].
– Reported by [Center for Statistical Science](#) and [School of Mathematical Sciences](#), Peking University.
5. Order Types and Natural Price Change: Model and Empirical Study of the Chinese Market
Siyu Liu, Chaoyi Zhao*, and Lan Wu.
International Journal of Financial Engineering, 9(04), 2250033, 2022. [[PDF](#)][[Journal](#)]

High-Dimensional Statistics, Machine Learning, and Artificial Intelligence

6. On Consistency of Signature Using Lasso
Xin Guo, Binnan Wang, Ruixun Zhang, and Chaoyi Zhao^{#*}.
Operations Research, 73(5), 2530–2549, 2025. [[PDF](#)][[Appendix](#)][[Journal](#)]
– Reported by [Center for Statistical Science](#), Peking University.
7. The Evolution of Discrimination Under Finite Memory Constraints
Andrew W. Lo, Ruixun Zhang, and Chaoyi Zhao^{#*}.
Scientific Reports, 15, 31774, 2025. [[PDF](#)] [[Appendix](#)] [[Journal](#)]

8. Interpretable Image-Based Deep Learning for Price Trend Prediction in ETF Markets
Ruixun Zhang, Chaoyi Zhao, and Guanglian Lin.
The European Journal of Finance, forthcoming, 2023. [[PDF](#)][[Journal](#)]
9. The Success of AdaBoost and its Application in Portfolio Management
Yijian Chuan, Chaoyi Zhao, Zhenrui He, and Lan Wu.
International Journal of Financial Engineering, 8(02), 2142001, 2021. [[PDF](#)][[Journal](#)]

Actuarial Sciences

10. The Checkerboard Copula and Dependence Concepts
Liyuan Lin, Ruodu Wang, Ruixun Zhang, and Chaoyi Zhao[#].
SIAM Journal on Financial Mathematics, 16(2), 426–447, 2025. [[PDF](#)][[Journal](#)]
11. Construct Smith–Wilson Risk-Free Interest Rate Curves with Endogenous and Positive Ultimate Forward Rates
Chaoyi Zhao^{*}, Zijian Jia, and Lan Wu.
Insurance: Mathematics and Economics, 114, 156–175, 2024. [[PDF](#)][[Journal](#)]

Books

12. Quantitative Investment (in Chinese)
Jian Sun, Lan Wu, and Chaoyi Zhao.
Science Press, 2023. [[Link](#)]
13. Financial Market Risk Management Analytics (in Chinese)
Frank H. Koger. Translated by Chaoyi Zhao.
Truth & Wisdom Press, 2022. [[Link](#)]
14. The Maths Book (in Chinese)
Dorling Kindersley Ltd. Translated by Chaoyi Zhao.
Publishing House of Electronics Industry, 2021. [[Link](#)]

Doctoral Dissertation

15. Stochastic Models and Statistical Analysis for High-dimensional Multi-factor Models and Induced Order Statistics
Chaoyi Zhao.
Doctoral Dissertation in Statistics, Peking University, 2024.

Preprints

16. No Asset Is an Island: A Real Options Framework for Valuing Interdependent Projects
Shomesh E. Chaudhuri, Andrew W. Lo, and Chaoyi Zhao^{#*}.
Under review.
17. Optimal Multi-Period Dynamic Portfolios for Sustainable Investing
Danping Li, Ruixun Zhang, and Chaoyi Zhao^{#*}.
Under review. [[SSRN](#)]
18. Underperformance of Performance Shares
Marc Hodak, Andrew W. Lo, Austin Starkweather, and Chaoyi Zhao[#].
Working paper. [[SSRN](#)]

19. Breaking the Factor Zootopia: Factor Selection via Significant Tests of Multi-Task Lasso
Chaoyi Zhao* and Lan Wu.
Working paper.

Research Projects

Data-driven Financial Risk Forecasting	2023 – 2024
National Key R&D Program of China, participant	
Portfolio Analytics and Risk Management for Green Investing: Theory and Application	2023 – 2024
National Natural Science Foundation of China, participant	
High-Frequency T+0 Quantitative Trading Strategy Based on Deep Learning	2019 – 2020
Peking University and Founder Securities, participant	
Gradient Optimization Model Based on Genetic Programming	2019
Ernst & Young (China) Advisor Ltd, intern	
Multi-Factor Alpha Strategy Based on the AdaBoost Algorithm	2018 – 2019
Peking University and Founder Securities, participant	
Application of Statistical Learning in Modeling the Relationship Between Stock Index and Futures	2017 – 2018
Peking University, undergraduate research, group leader	

Teaching Experience

Teaching Assistant at Massachusetts Institute of Technology	
<i>15.C51 Modeling with Machine Learning: Financial Technology</i>	Spring 2026
Elective for Graduate Students. Instructors: Prof. Andrew W. Lo and Prof. Paul F. Mende	
<i>15.482 Healthcare Finance</i>	Fall 2024
Elective for Graduate and Senior Undergraduate Students. Instructor: Prof. Andrew W. Lo	
Teaching Assistant at Peking University	
<i>Quantitative Trading</i>	Fall 2020, 2022
Elective for Graduate and Senior Undergraduate Students. Instructors: Prof. Jian Sun and Lan Wu	
<i>Practice for Fixed Income Securities</i>	Fall 2021, 2022
Elective for Graduate and Senior Undergraduate Students. Instructors: Prof. Lan Wu <i>et al.</i>	
<i>Theory and Practice for Futures and Other Derivatives</i>	Spring 2022
Elective for Graduate and Senior Undergraduate Students. Instructors: Prof. Lan Wu <i>et al.</i>	
<i>Mathematical Analysis III (Including Exercise Class)</i>	Fall 2021
Core Class for Undergraduate Students. Instructor: Prof. Jiazhong Yang	
<i>Mathematical Analysis II (Including Exercise Class)</i>	Spring 2021
Core Class for Undergraduate Students. Instructor: Prof. Jiazhong Yang	
<i>Risk Management and Financial Regulation</i>	Spring 2021
Core Class for Graduate Students. Instructor: Prof. Lan Wu	
<i>Exercise Class for Probability and Statistics</i>	Spring 2020
Elective for Undergraduate Students.	
<i>Introduction to Financial Mathematics</i>	Fall 2019
Core Class for Undergraduate Students. Instructor: Prof. Lan Wu	

Talks

No Asset Is an Island: A Real Options Framework for Valuing Interdependent Projects

Peking University Seminar

Sep 18, 2025

On Consistency of Signature Using Lasso

SIAM Conference on Financial Mathematics and Engineering (FM25)

Jul 15, 2025

(Session Chair) 2024 INFORMS Annual Meeting

Oct 22, 2024

CSIAM 2023

Oct 15, 2023

UC Berkeley Seminar

May 7, 2023

Breaking the Factor Zootopia: Factor Selection via Significant Tests of Multi-Task Lasso

7th PKU-NUS Annual International Conference on Quantitative Finance and Economics

May 20, 2023

Peking University Seminar

Apr 21, 2023

CSIAM Financial Math Annual Conference

Nov 13, 2022

Optimal Impact Portfolios with General Dependence and Marginals

ICIAM 2023

Aug 20–25, 2023

The 11th Annual Conference of the Operations Research Society of China (Financial Engineering and Risk Management Branch)

Dec 10, 2022

– Second place in the Best Paper Prize for Young Scholars

Sun Yat-sen University Seminar

Nov 4, 2022

The 19th Chinese Finance Annual Meeting

Oct 30, 2022

High-Frequency Liquidity in the Chinese Stock Market: Measurements, Patterns, and Determinants

China Finance Review International & China International Risk Forum Joint Conference

Jul 29, 2023

CSIAM Financial Math Annual Conference

Nov 12, 2022

IFZ FinTech Colloquium 2022, Lucerne University of Applied Sciences and Arts

Nov 2, 2022

2022 INFORMS Annual Meeting

Oct 17, 2022

(Session Chair) 11th World Congress of Bachelier Finance Society

Jun 15, 2022

Construct Smith–Wilson Interest Rate Curves with Endogenous Ultimate Forward Rates

United As One: 24th International Congress on Insurance: Mathematics and Economics

Jun 5, 2021

Statistical Patterns of High-Frequency Data for A-share stocks in Shenzhen Stock Exchange

3rd Seminar for Young Scholars in Financial Engineering and Risk Management

Jun 12, 2021

Discussions

2025 MRS International Risk Conference

July 26, 2025

Non-Academic Talks

Speech for the Opening Ceremony of School of Mathematical Sciences, Peking University

Sep 15, 2018

– Representative of seniors [[Link](#)]

Academic Services

Referee Service

ASTIN Bulletin, IEEE Transactions on Circuits and Systems for Video Technology, IEEE Transactions on Multimedia, Insurance: Mathematics and Economics, Journal of Financial Data Science, Management Science, North American Actuarial Journal, Operations Research

Skills

Programming: Python, R, C/C++, MATLAB, Java, SQL

Languages: Mandarin Chinese, English

Hobbies: Swimming, Rowing, Hiking, Bouldering, Stand-up Comedy